



A DATA-DRIVEN ANALYSIS OF UNEMPLOYMENT AND ITS DETERMINANTS IN JORDAN

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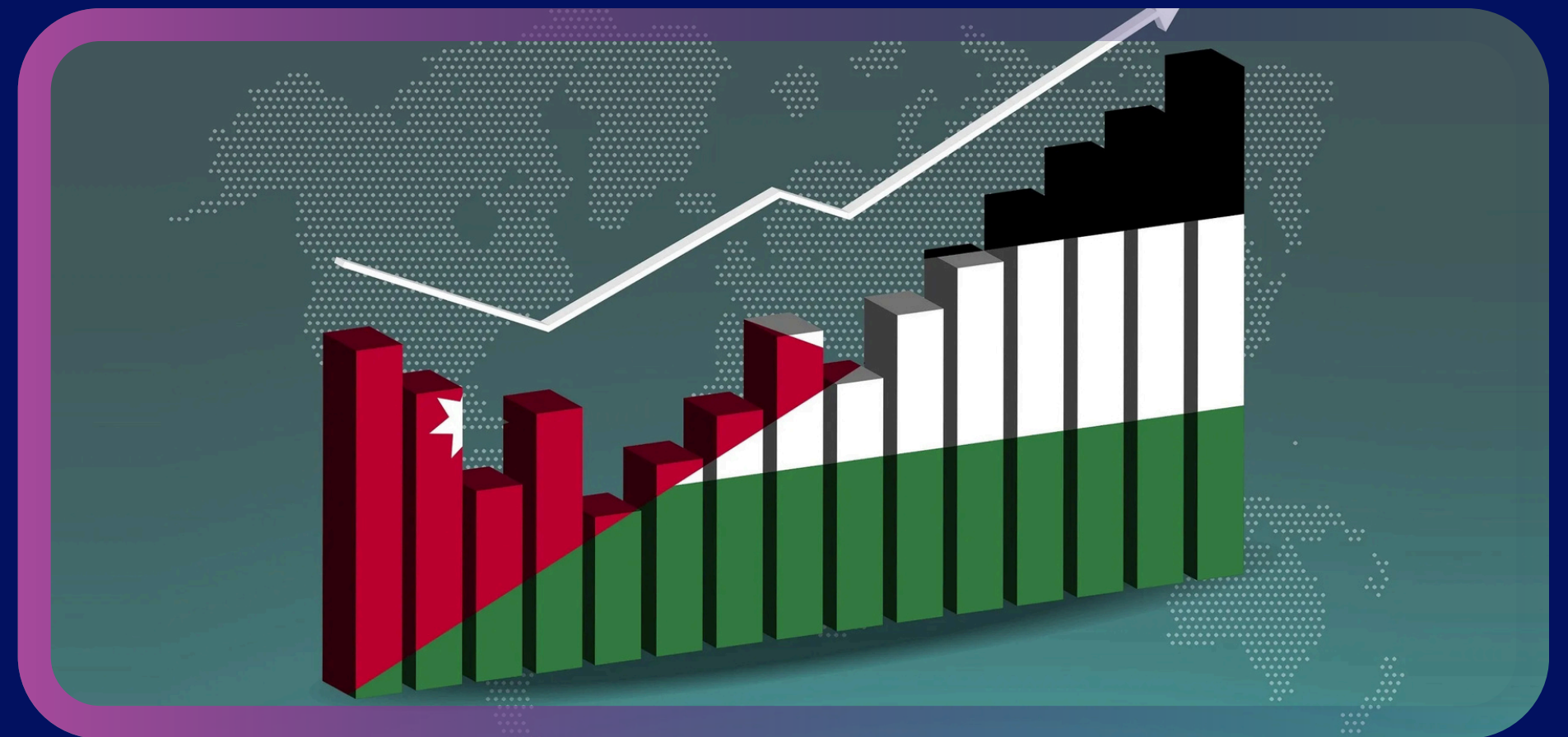
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JORDAN'S ECONOMIC CONTEXT

- Jordan depends strongly on human capital, education, services, tourism, and trade.
- The economy faces limited natural resources, population growth, and job creation pressure.
- Unemployment is one of the major labor market challenges.
- Understanding unemployment requires looking at both time and regional differences





WHY WE CHOSE THIS PROJECT ?

- The idea started from a real-life observation.
- While searching for a graduation project topic, we heard fresh graduates discussing job difficulties.
- Some graduates were considering working outside Jordan because they could not find opportunities in their fields.
- This made us think about unemployment as a real issue affecting youth and graduates.
- Dr. Husam Barham encouraged us to study unemployment across all 12 Jordanian governorates.





FROM OBSERVATION TO RESEARCH QUESTION

01 Real-life observation

02 Graduates struggling to find jobs

03 Unemployment differs across governorates

04 Possible economic, demographic, and educational factors

05 Data-driven analysis across 12 governorates





WHY UNEMPLOYMENT MATTERS?





HOW WE THOUGHT ABOUT THE PROBLEM?

Dimension	Variable Example	Why It Matters
Economic activity	Establishments per 1,000 people aged 15+	More establishments may create more job opportunities
Demographic pressure	Population Density	Higher density may increase competition for jobs
Education infrastructure	Schools per 1,000 population	Education affects skills and employability





PROJECT OBJECTIVE

The main objective is to analyze unemployment in Jordan using a data-driven approach.

Specific objectives:

- Analyze unemployment trends from 2014 to 2024.
- Study unemployment across all 12 governorates.
- Identify socioeconomic factors related to unemployment.
- Compare governorates using dashboards and clustering.
- Support data-driven decision-making.





Main Dataset Variables

Variable	Meaning
Unemployment Rate	Main target variable
Population Density	Labor market pressure indicator
Establishments per 1,000	Economic activity indicator
Schools per 1,000 Population	Education infrastructure indicator
Students per Class	Education capacity indicator
Student Ratio 5–19	Education participation indicator

Data Sources

- Jordan Department of Statistics: unemployment, employment, labor force, population.
- Social Security Corporation: number of establishments.
- Ministry of Education: schools, students, classes, education indicators.





Panel Data Structure

Panel Data = Cross-sectional data + Time-series data

In this project, the dataset combines:

1. Cross-sectional dimension

Jordanian governorates

Example: Amman, Irbid, Zarqa, Aqaba, Ma'an, Mafraq...

2. Time dimension

Years from 2014 to 2024

Each row represents:

One governorate observed in one specific year



Why Panel Data?

- Tracks unemployment changes over time.
- Compares unemployment across governorates.
- Supports Fixed Effects modeling.
- Captures regional differences more accurately.



Data Preparation & Derived Indicators

Data preparation included:

- Renaming columns
- Converting data types
- Handling missing values
- Selecting relevant features
- Standardizing variables for clustering
- Exporting cleaned outputs

For 2014–2016, some population age data was available as percentages, so actual population values were calculated using:

$$\text{Total Population} \times \text{Population Rate} / 100$$





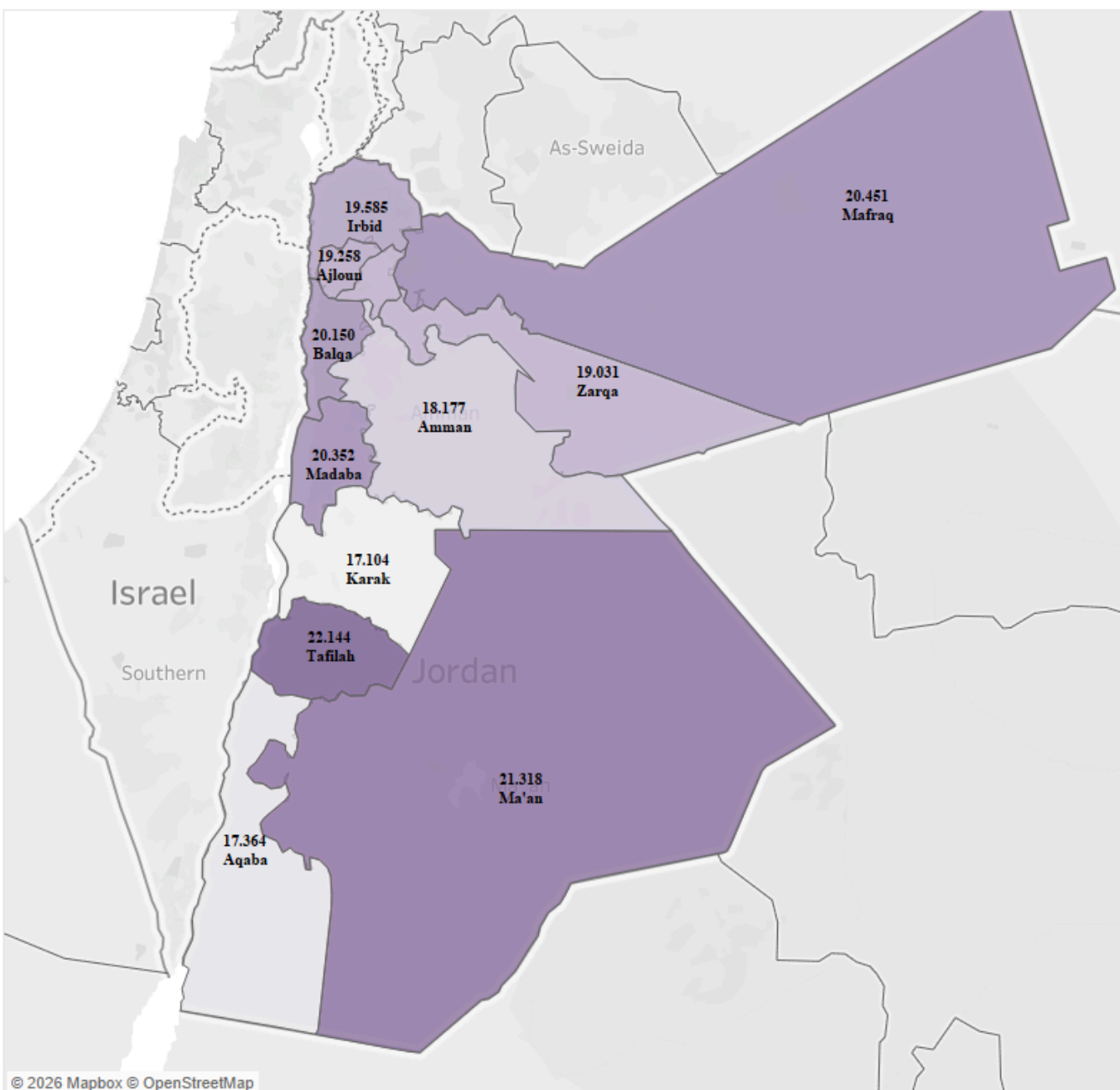
WHAT IS HAPPENING IN JORDAN'S LABOR MARKET?

Dashboard 1: Labor Market Overview

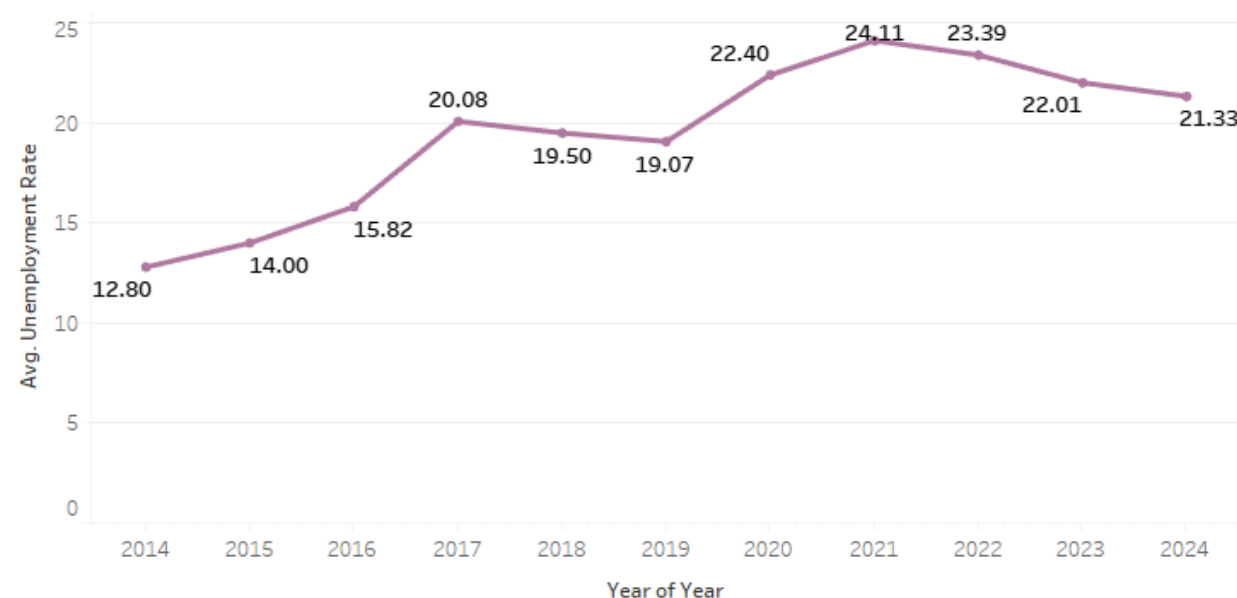


Labor Market Overview

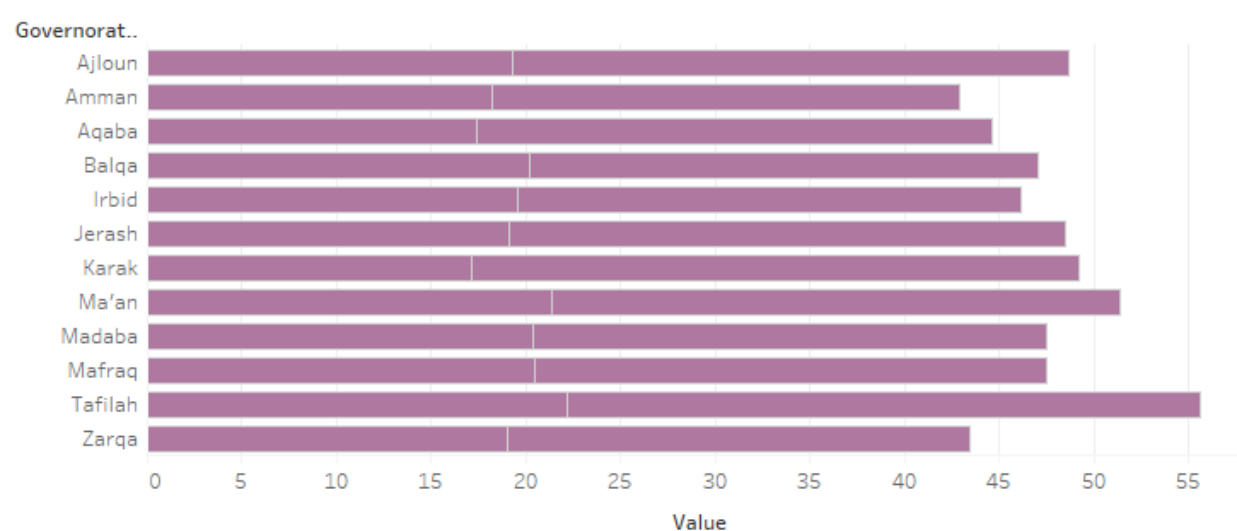
Unemployment Rate by Governorate in Jordan



Unemployment Trend by Year



Employments Rate vs Unemployments Rate



What this dashboard answers

Where is unemployment spatially concentrated?
How did unemployment change from 2014 to 2024?
How do employment and unemployment profiles compare across governorates?

12.8

avg. unemployment in 2014

24.1

peak in 2021

21.3

avg. unemployment in 2024

Main reading

Unemployment rises strongly after 2016 and peaks around 2021.
The map highlights variation across governorates rather than one uniform national pattern.
The employment/unemployment view supports a fast labor-market comparison by governorate.

Dashboard role

This dashboard tells the “what is happening?” part of the story before moving to explanatory factors.





EXPLORING POSSIBLE FACTORS

Dashboard 2: Key Determinants

Dashboard structure

Three scatter plots show bivariate visual relationships.
Trend lines indicate the direction of each relationship.
The coefficient chart summarizes the final model results.

Visual patterns

Establishment density: weak negative visual relationship.
Population density: very weak negative visual relationship.
Schools per 1,000: visible positive bivariate trend in the dashboard.

Model-based interpretation

Strongest coefficient by absolute value: Schools per 1,000 population (-0.1262).
Lightest effect: Population density (-0.0033).
Establishments per 1,000 shows a small positive coefficient ($+0.0500$).

Important note

Scatter plots are descriptive; coefficients provide the model-based evidence.





Why Modeling Was Needed?

- Dashboards show visual patterns.
- Statistical models measure relationships more formally.
- Pooled OLS was used as a baseline.
- Fixed Effects was used because governorates are different.
- LOYO was used to test generalization across years.





VIF ANALYSIS

Variance Inflation Factor was used to check whether independent variables were highly correlated.

Interpretation:

- VIF 1–5: Acceptable
- VIF 5–10: Moderate
- VIF > 10: High multicollinearity

The selected regression variables were:

- Establishments per 1,000 people aged 15+
- Population Density
- Schools per 1,000 Population

The selected regression variables showed acceptable VIF levels, indicating no severe multicollinearity problem.

$$VIF_i = \frac{1}{1 - R_i^2}$$





Modeling Techniques

The project used regression models, validation methods, and clustering techniques to analyze unemployment patterns and socioeconomic determinants.

Pooled OLS Model

Pooled OLS was used as the baseline regression model

$$R^2 = 0.281$$
$$RMSE = 3.586$$
$$MAE = 2.955$$

The model captured general patterns but had limited explanatory power.

Fixed Effects Model

Fixed Effects was used because the dataset is panel data and governorates have different characteristics.

$$R^2 = 0.847$$
$$RMSE = 1.657$$
$$MAE = 1.306$$

Fixed Effects achieved the best performance by controlling for governorate-specific differences.

LOYO Validation

LOYO validation was used to test temporal generalization

$$\text{Overall } R^2 = 0.692$$

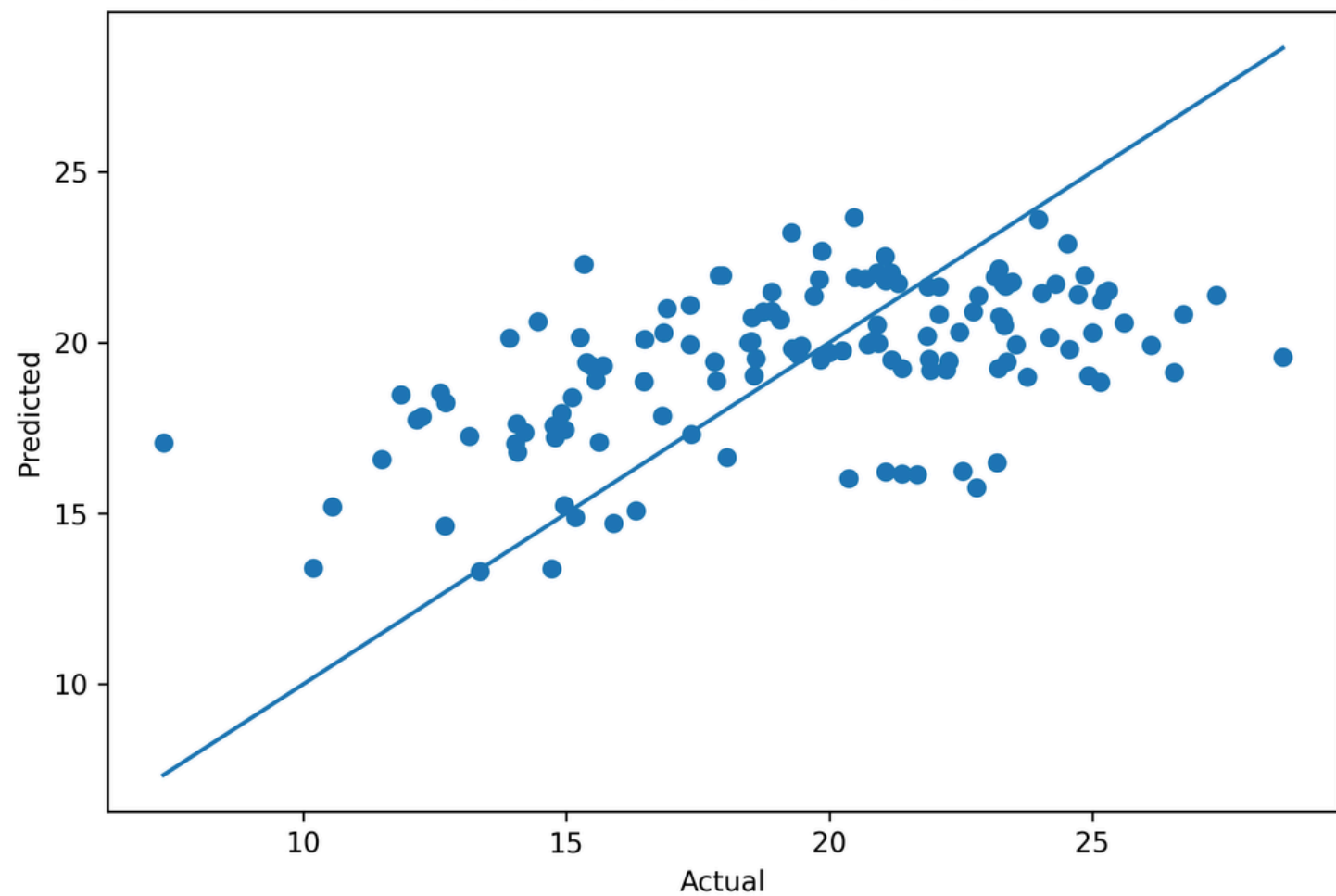
Process:

One year is removed → Model trains on remaining years → Model predicts excluded year

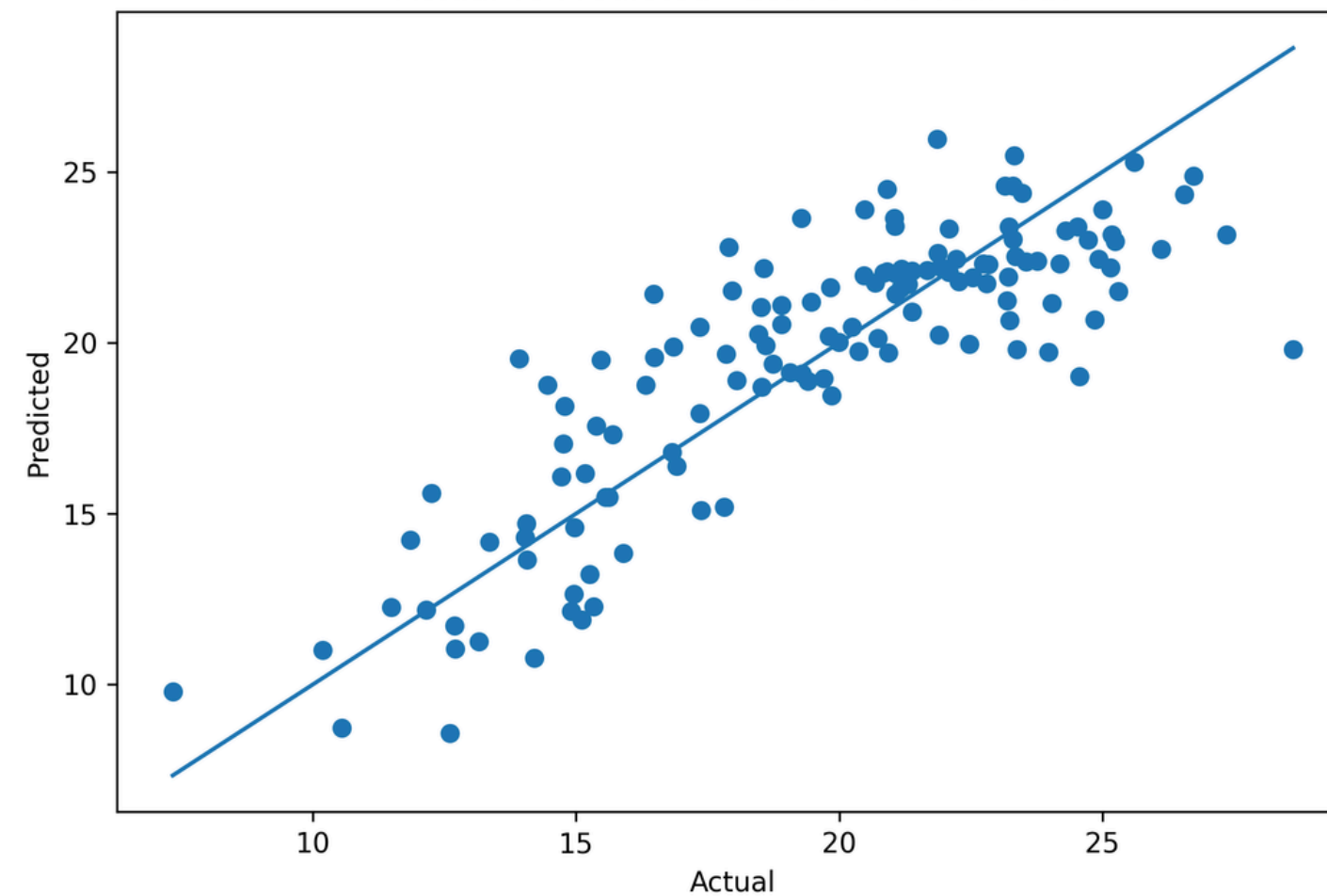
The LOYO validation showed that the model had acceptable temporal generalization and was not only fitting the training data.



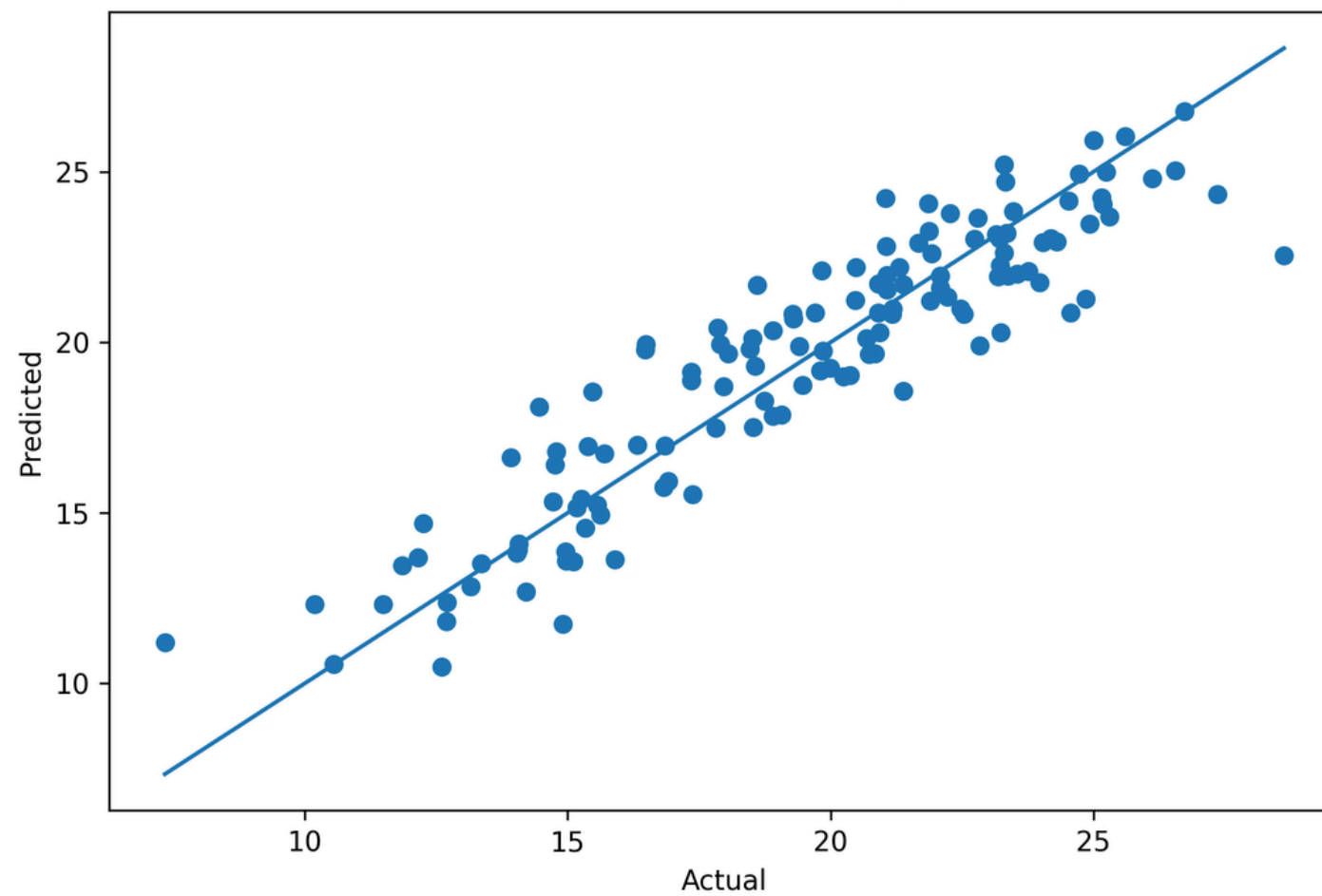
Actual vs Predicted - Pooled OLS



Actual vs Predicted - Leave-One-Year-Out



Actual vs Predicted - Fixed Effects





CLUSTERING ANALYSIS & PCA

VISUALIZATION

Clustering was used to group Jordanian governorates based on socioeconomic similarity, while PCA was applied to reduce the data dimensions and visualize the clusters clearly.

1. Clustering Features

The clustering model used several indicators:

Unemployment Rate

Establishments per 1,000 people aged 15+

Population Density

Students per Class

Schools per 1,000 Population

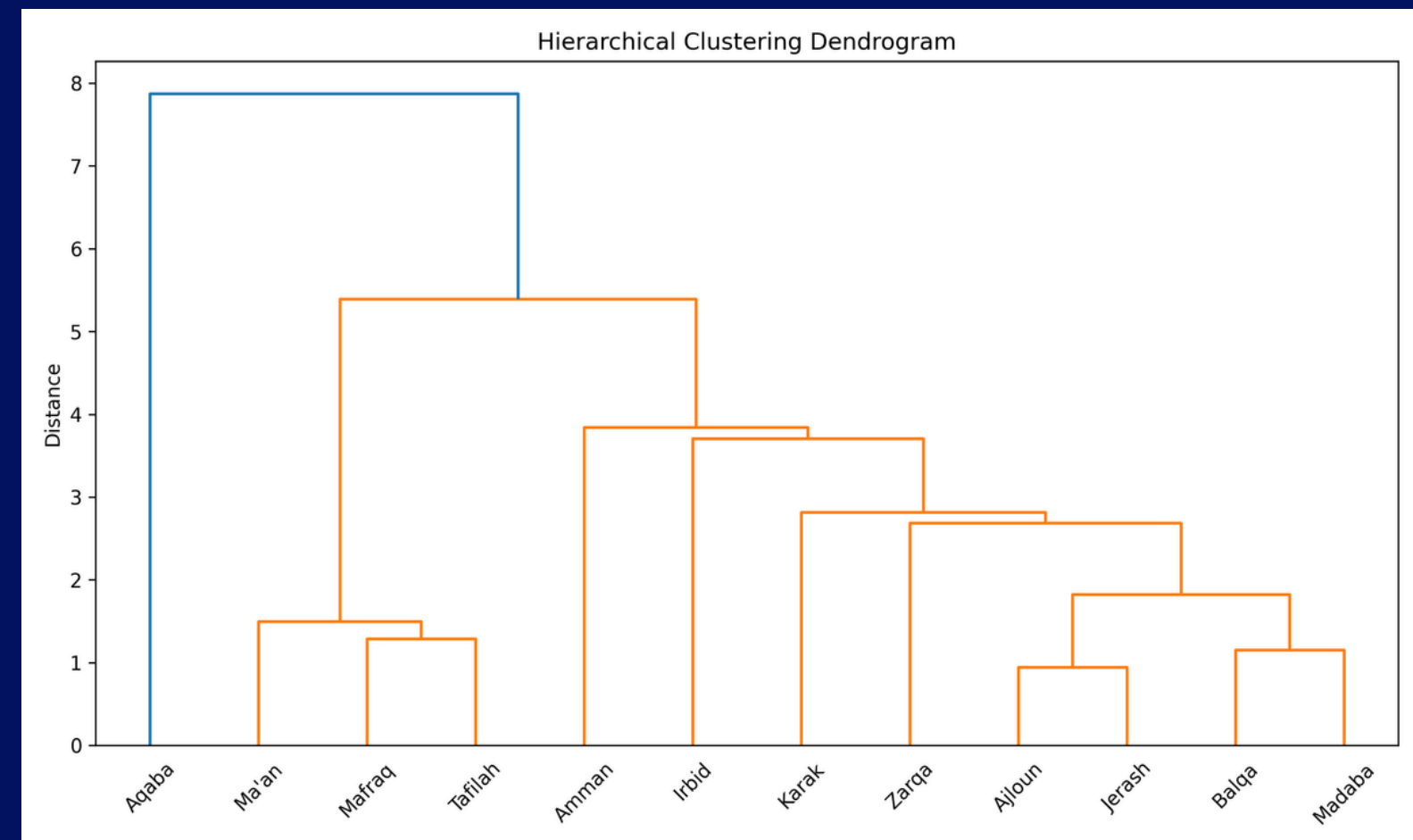
Student Ratio 5–19

2. Methods Used

Two clustering techniques were applied:

K-Means Clustering: grouped governorates into clusters based on similarity.

Hierarchical Clustering: showed the similarity structure between governorates using a dendrogram.





CLUSTERING ANALYSIS & PCA VISUALIZATION

3. PCA Role

PCA was used to transform multiple variables into two principal components:

$$PC_1 = a_1X_1 + a_2X_2 + a_3X_3 + \dots + a_nX_n$$

This made it possible to visualize governorates in a 2D space while preserving the most important variance in the data.

4. Validation Results

The optimal number of clusters was selected using:

Elbow Method

Silhouette Score

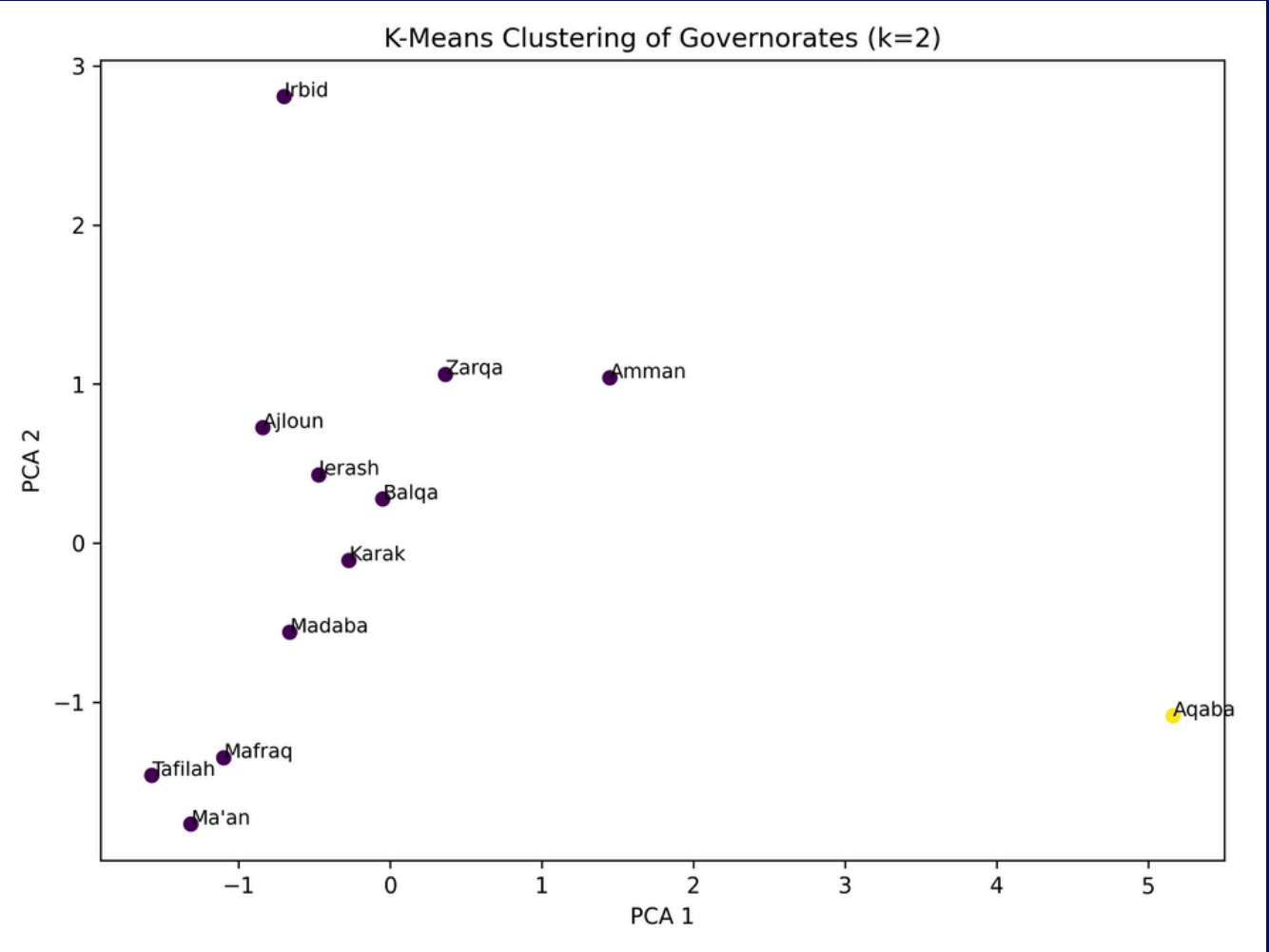
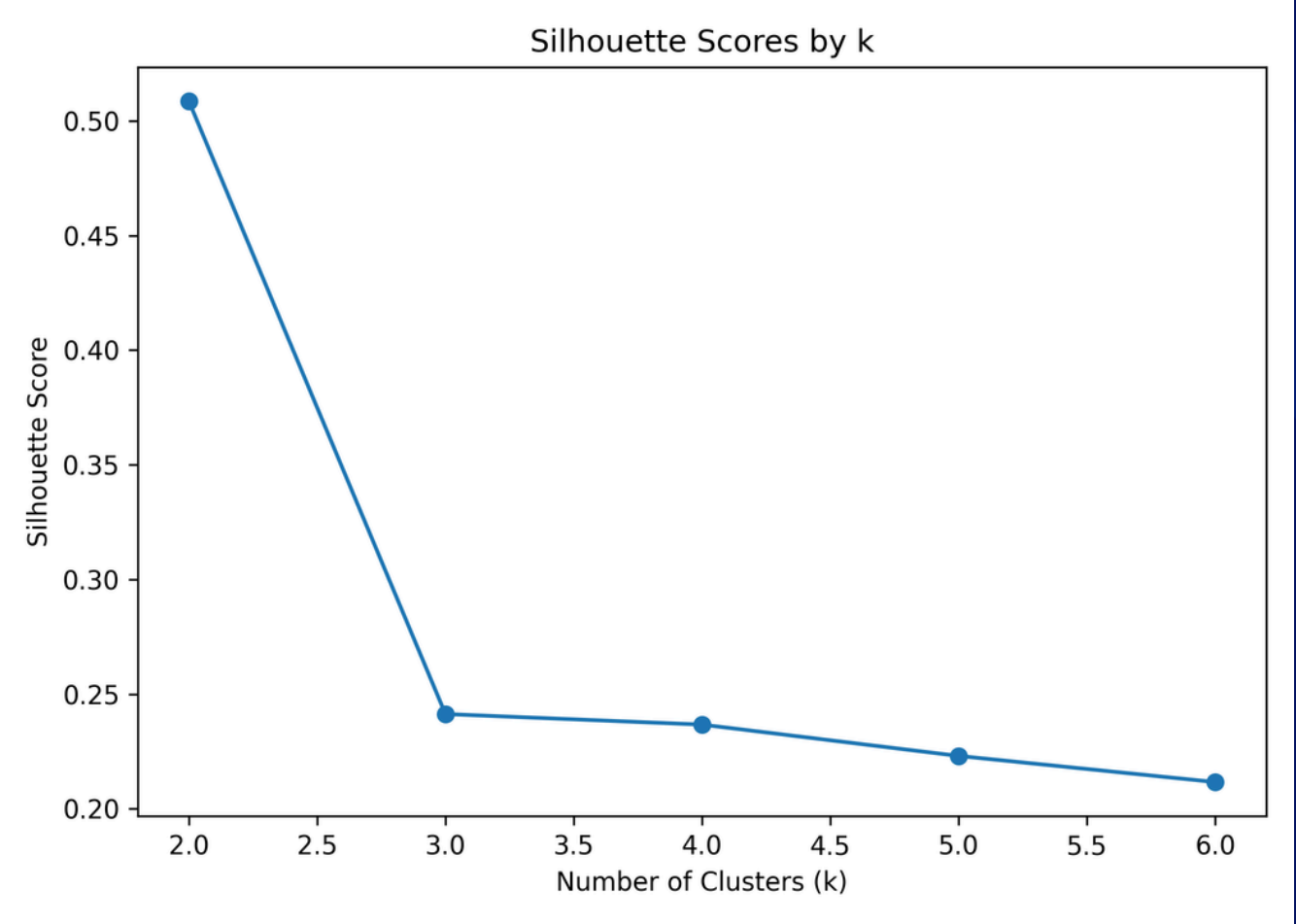
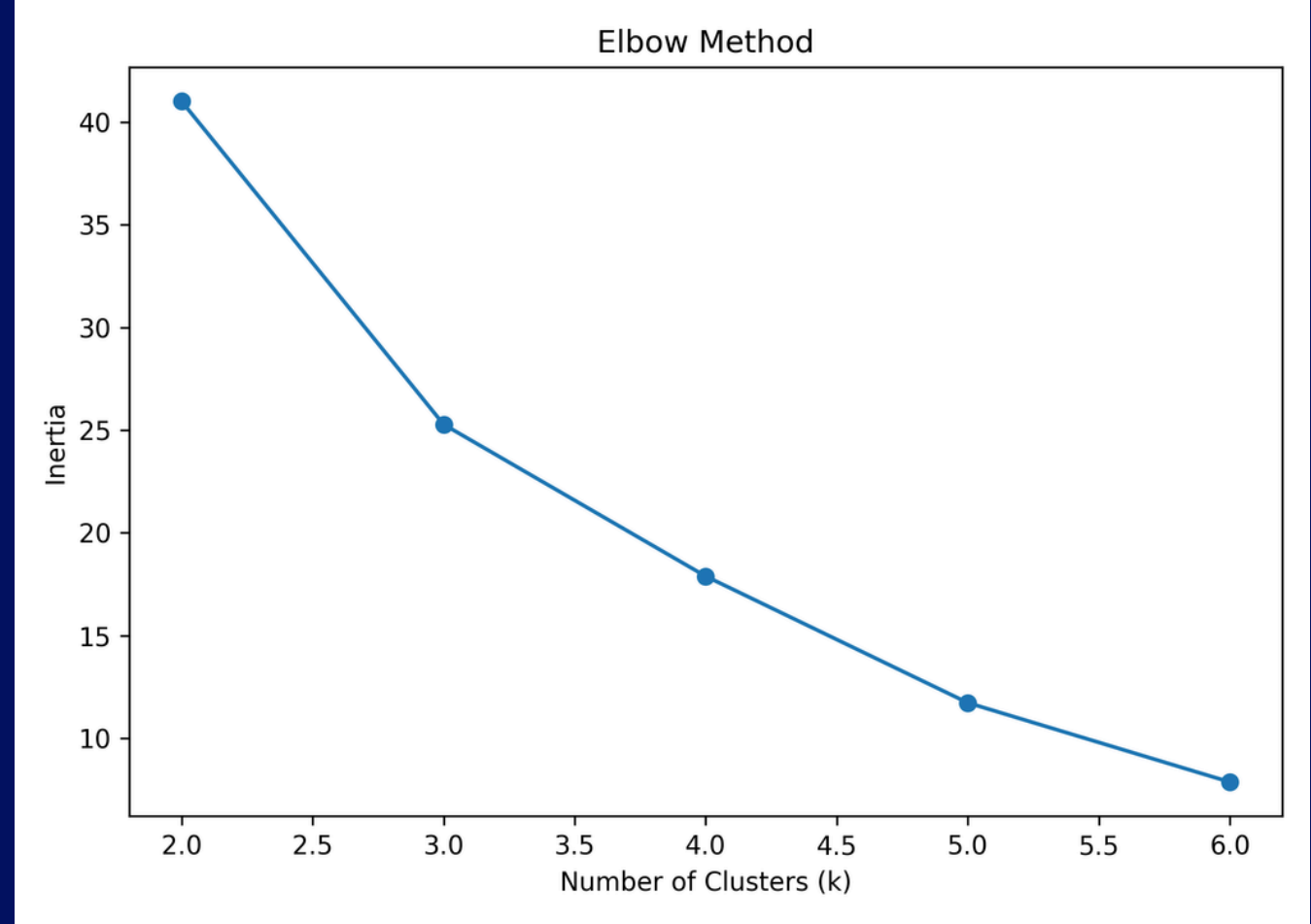
Final Result:

Best number of clusters = 2

Key Insight

The clustering results showed that Jordanian governorates can be grouped into two main socioeconomic patterns. PCA helped visualize these groups clearly, and Hierarchical Clustering confirmed the similarity relationships between governorates.







CONCLUSION & RECOMMENDATIONS

I. What We Found from the Data?

- Unemployment varies clearly across Jordanian governorates.
- Fixed Effects showed that regional differences are important.
- Economic activity, population density, and education indicators are related to unemployment.
- Clustering revealed two main socioeconomic patterns among governorates.

Recommendations

- Encourage investment in high-unemployment areas.
- Improve alignment between education and labor market needs.
- Create governorate-specific employment policies.
- Use BI dashboards for continuous monitoring and decision-making





THANK YOU FOR YOUR ATTENTION

ANY QUESTIONS ?

